MINI PROJECT – GENERATIVE AI

AI PERSONAL STUDY ASSISTANT

Study Material: Generative Artificial Intelligence

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Purpose: To create an easy-to-use study resource that explains Generative AI and supports learning through questions, examples, and revision activities.

1. Introduction to Generative AI

Generative Artificial Intelligence (Generative AI) is a type of AI that can create new content based on patterns learned from examples. It can generate text, images, audio, video, and computer code.

Examples of Generative AI tools

• ChatGPT – generates and explains text.
• Image-generation tools – create pictures from text prompts.
• AI coding assistants – suggest or explain code.
• Speech and music tools – create or transform audio.

How it works (simple view)

1. Training: the model learns patterns from large collections of data.
2. Prompt: a user gives an instruction or question.
3. Generation: the model predicts suitable next pieces of content.
4. Review: the user checks the output and improves the prompt if needed.

Important note

Generative AI does not think or understand exactly like a human. It produces outputs using learned patterns and the context supplied to it. Its answers can be useful, but they can also be incorrect or incomplete.

Key terms

AI: technology that performs tasks associated with human intelligence.
Model: a system trained to identify patterns and produce outputs.
Prompt: the instruction or input given to an AI tool.
Output: the content produced by the tool.

2. Generative AI vs Traditional AI

Feature	Traditional AI	Generative AI
Main purpose	Classify, predict, or decide	Create new content
Typical output	A label, score, or prediction	Text, image, audio, code, etc.
Example	Spam email detection	Drafting an email
Study use	Predicting performance from data	Explaining a topic or making a quiz

Common applications

• Education: summaries, practice questions, explanations, and study plans.
• Healthcare: drafting and summarising information with professional review.
• Business: creating first drafts of reports, emails, and marketing ideas.
• Software: explaining code, generating examples, and helping find bugs.
• Creative work: brainstorming stories, designs, and presentations.

Benefits

1. Can save time on first drafts.
2. Can explain difficult ideas in different ways.
3. Can support personalised practice.
4. Can help people brainstorm and explore ideas.

Limitations and risks

1. May produce false or made-up information.
2. May reflect bias in its training data.
3. May not know the latest facts.
4. Can raise privacy, copyright, and academic-integrity concerns.
5. Human checking is important, especially for important decisions.

3. Prompt Engineering

Prompt engineering means writing clear instructions to guide an AI system toward a useful response.

Parts of a good prompt

• Role: tell the AI what perspective to use.
• Task: state exactly what it should do.
• Context: provide the topic, audience, and purpose.
• Format: specify bullets, a table, steps, or a paragraph.
• Constraints: mention length, language, level, and exclusions.

Weak prompt vs improved prompt

Weak: “Explain AI.”
Improved: “Explain Generative AI to a first-year engineering student in simple English. Give a definition, five applications, three benefits, three limitations, and five short revision questions. Use headings and bullet points.”

Prompt improvement checklist

1. Is the task specific?
2. Is the audience clear?
3. Did I give enough context?
4. Did I request the format I need?
5. Did I ask for examples or a step-by-step explanation where useful?
6. Have I checked the answer for accuracy?

Practice prompt

“Act as a friendly study tutor. Explain large language models in simple English using one everyday analogy. Then give three key points and two quiz questions with answers.”

4. How an AI Personal Study Assistant Works

An AI Personal Study Assistant helps learners understand a topic, review notes, and practise questions. In this project, the study material is stored as a PDF and a chatbot uses that document as a knowledge source.

Basic workflow

1. Prepare: write clear study notes and save them as a PDF.
2. Build: upload the PDF to a document-question-answering chatbot platform, such as Chatbase.
3. Configure: give the bot instructions to answer from the document and admit when the answer is not available.
4. Test: ask questions whose answers are in the PDF and questions that are outside it.
5. Publish: create a simple website with a project introduction, study topics, and the chatbot embed or link.

Suggested website sections

• Home: project title and short introduction.
• Learn: key topics and a link to the PDF.
• Ask the AI: chatbot embed or link.
• How to use: examples of useful questions.
• Limitations: reminder to verify important answers.

Sample chatbot instruction

“You are an AI Personal Study Assistant for students learning Generative AI. Answer using the uploaded study material whenever possible. Explain in simple English, use short headings and examples, and help the learner revise. If the document does not contain the answer, say so clearly instead of inventing information. Encourage students to verify important facts.”

Example questions to ask

• What is Generative AI?
• Give three differences between traditional AI and Generative AI.
• What is a prompt?
• List four risks of Generative AI.
• Explain the workflow of this study assistant.

5. Responsible and Effective Use

Generative AI is most useful when it supports learning rather than replacing the learner's own thinking.

Good practices

1. Verify facts using textbooks, class notes, or reliable sources.
2. Do not enter passwords, private records, or sensitive personal information.
3. Follow your college's rules on AI use and academic honesty.
4. Use AI explanations to understand a concept, then write the final answer in your own words.
5. Check calculations, quotations, references, and code before relying on them.
6. Remember that an answer can sound confident and still be wrong.

A simple study routine

- Learn: ask for a short explanation of one topic.
- Review: ask for five key points.
- Practise: answer quiz questions without looking at notes.
- Check: ask for feedback and review mistakes.
- Remember: summarise the topic in your own words.

Project evaluation checklist

- ■ PDF has a title page and at least 5–8 pages.
- ■ Text is readable and headings are consistent.
- ■ Chatbot has the correct PDF uploaded as its knowledge source.
- ■ Chatbot answers document-based questions correctly.
- ■ Website clearly explains the project and provides access to the chatbot and PDF.
- ■ Limitations and responsible-use advice are visible.

6. Revision Questions and Answers

1. What is Generative AI?
Generative AI is a type of AI that creates new content, such as text, images, audio, video, or code, based on patterns learned from data.

2. What is a prompt?
A prompt is the instruction, question, or input given to an AI system.

3. Give two examples of Generative AI uses.
Examples include generating a text summary and creating an image from a written description.

4. State two differences between traditional AI and Generative AI.
Traditional AI often classifies or predicts; Generative AI creates content. Traditional AI may output a label or score, while Generative AI may output text, images, audio, or code.

5. What is prompt engineering?
It is the practice of writing and improving instructions so that an AI tool gives a more useful response.

6. Mention three limitations of Generative AI.
It may give incorrect information, reflect bias, and raise privacy or copyright concerns.

7. Why should AI answers be checked?
AI can generate plausible-sounding but incorrect or incomplete information, so important claims need verification.

8. What is the purpose of the project chatbot?
It answers student questions using the uploaded Generative AI study PDF as a knowledge source.

9. What should the chatbot do if the answer is not in the PDF?
It should clearly say that the information is not available in the provided material instead of making up an answer.

10. Give two responsible-use practices.
Verify important facts and avoid entering sensitive personal information.

Short practice activity

Write a prompt asking an AI tutor to explain a topic from your course. Include the audience, the required format, and at least one constraint. Then review the response for accuracy and clarity.

7. Summary and Glossary

Summary

• Generative AI creates content by learning patterns from data.
• Prompts guide the system's response; clear prompts usually make the task easier to understand.
• Generative AI can support study through explanations, summaries, and quizzes.
• Outputs may be wrong or biased, so students must check them.
• A document-based chatbot can answer questions from a supplied PDF, but it may still misunderstand questions or retrieve incomplete information.
• A useful study assistant should be simple, transparent, and designed to support independent learning.

Glossary

Artificial Intelligence (AI): computer systems performing tasks that usually require human-like abilities.
Generative AI: AI that produces new content.
Prompt: an instruction or input to an AI system.
Prompt engineering: improving prompts to guide outputs.
Large Language Model (LLM): a model trained on text and other data to process and generate language.
Chatbot: a system that communicates with users through conversation.
Knowledge source: information a chatbot can use when answering questions.
Hallucination: an AI-generated claim that is false or unsupported.
Bias: a systematic tendency that can lead to unfair or unbalanced outputs.

End of Study Material

Prepared for the IBM Mini Project – Generative AI: AI Personal Study Assistant.